Acceptance statistics (statistics)

Statements digest, version 0.10.0

Every definition, fact, assumption, result and requirement of the contract, with the contract's numbers; remarks, examples, technical steps and all proofs are left out. Each heading opens its statement in the contract.

Mathematics

1 Frequencies and acceptance geometries

Definition 1.1 (Frequency vector and counts)**.** For an outcome string $\hat{c}_1^n = (\hat{c}_1, \dots, \hat{c}_n) \in \widehat{\mathcal{C}}^n$ and $\hat{c} \in \widehat{\mathcal{C}}$ put

$$\mathsf{N}_{\hat{c}}(\hat{c}_1^n) := \big|\{\, j \in \{1, \dots, n\} : \hat{c}_j = \hat{c} \,\}\big|, \qquad \mathrm{freq}_{\hat{c}_1^n}(\hat{c}) := \tfrac{1}{n}\, \mathsf{N}_{\hat{c}}(\hat{c}_1^n) : \tag{1.1}$$

the *count* of the outcome $\hat{c}$ and the *frequency vector* of the string. Then $\mathrm{freq}_{\hat{c}_1^n} \in \Delta(\widehat{\mathcal{C}})$ for every string, and its coordinates lie in $\frac{1}{n}\{0, 1, \dots, n\}$.

Lemma 1.2 (Frequency vectors in the probability simplex)**.** Let $n \geq 1$, and write $\Delta(\widehat{\mathcal{C}})$ for the set of $\mathbf{q} \in \mathbb{R}^{\widehat{\mathcal{C}}}$ with non-negative coordinates summing to one.

(i) For every outcome string $\hat{c}_1^n \in \widehat{\mathcal{C}}^n$, every count $\mathsf{N}_{\hat{c}}(\hat{c}_1^n)$ of Definition 1.1 is a non-negative integer, at most n of them are non-zero, and $\sum_{\hat{c} \in \widehat{\mathcal{C}}} \mathsf{N}_{\hat{c}}(\hat{c}_1^n) = n$. Hence $\mathrm{freq}_{\hat{c}_1^n} \in \Delta(\widehat{\mathcal{C}})$.

(ii) For every set $S \subseteq \mathbb{R}^{\widehat{\mathcal{C}}}$ and every random string $\widehat{C}_1^n$ with values in $\widehat{\mathcal{C}}^n$, the sets $\{\mathrm{freq}_{\widehat{C}_1^n} \in S \cap \Delta(\widehat{\mathcal{C}})\}$ and $\{\mathrm{freq}_{\widehat{C}_1^n} \in S\}$ of sample points are equal. In particular, whenever either is an event,

$$\Pr\big[\mathrm{freq}_{\widehat{C}_1^n} \in S \cap \Delta(\widehat{\mathcal{C}})\big] = \Pr\big[\mathrm{freq}_{\widehat{C}_1^n} \in S\big]. \tag{1.2}$$

Definition 1.3 (Alphabet and round count matched to the consuming declaration)**.** An acceptance alphabet $\widehat{\mathcal{C}}$ and a round count n are a *matched alphabet and round count* when $\widehat{\mathcal{C}}$ is the finite, non-empty acceptance alphabet of the same declaration $\mathfrak{D}$ (Definition "Protocol declaration" (protocol-compiler/def:declaration)) the consumer solves, the codomain of ϕ, with a fixed enumeration, and n is the same total round count.

Definition 1.4 (Admissible geometries and the canonical box form)**.** An *admissible geometry* is a pair consisting of a *tag* and its *data*, drawn from exactly the following two cases, each with its data, the set Γ the data denote and its *validation predicate*. There is no third tag.

singleton**.** Data: a vector $\mathbf{q}^* \in \mathbb{Q}^{\widehat{\mathcal{C}}}$. Denotation $\Gamma \coloneqq \{\mathbf{q}^*\}$. *Validation predicate:* $q^*(\hat{c}) \geq 0$ for every $\hat{c}$ and $\sum_{\hat{c}} q^*(\hat{c}) = 1$.

polytope**.** Data: an integer row count $m_{\mathrm{row}} \geq 0$, a matrix $B \in \mathbb{Q}^{m_{\mathrm{row}} \times \widehat{\mathcal{C}}}$ and a vector $\mathbf{b} \in \mathbb{Q}^{m_{\mathrm{row}}}$. Denotation

$$\Gamma \coloneqq \{\mathbf{q} \in \Delta(\widehat{\mathcal{C}}) : \; B\mathbf{q} \leq \mathbf{b}\}, \tag{1.3}$$

the inequality componentwise. *Validation predicate:* feasibility of the linear system $\{\mathbf{q} \geq 0,\ \sum_{\hat{c}} q(\hat{c}) = 1,\ B\mathbf{q} \leq \mathbf{b}\}$ over the rationals.

A polytope geometry is in *canonical box form* when it carries, in addition, a pair $\mathbf{q}^-, \mathbf{q}^+ \in \mathbb{Q}^{\widehat{\mathcal{C}}}$ with $0 \leq \mathbf{q}^- \leq \mathbf{q}^+$ componentwise, and its $(B, \mathbf{b})$ are exactly

$$m_{\mathrm{row}} = 2m, \qquad \begin{array}{lll} \text{row } i: & \delta_{\hat{c}^{(i)}}^{\mathsf{T}}, & \text{right-hand side } q^+(\hat{c}^{(i)}), \\ \text{row } m+i: & -\delta_{\hat{c}^{(i)}}^{\mathsf{T}}, & \text{right-hand side } -q^-(\hat{c}^{(i)}), \end{array} \qquad i = 1, \ldots, m, \tag{1.4}$$

where $m \coloneqq |\widehat{\mathcal{C}}|$, $\hat{c}^{(1)}, \ldots, \hat{c}^{(m)}$ is the declaration's enumeration of $\widehat{\mathcal{C}}$ and $\delta_{\hat{c}}$ is the standard basis vector of $\mathbb{R}^{\widehat{\mathcal{C}}}$ at $\hat{c}$. Row indices are integers, never alphabet elements; equivalently

$$\Gamma = \{\mathbf{q} \in \Delta(\widehat{\mathcal{C}}) : \; q^-(\hat{c}) \leq q(\hat{c}) \leq q^+(\hat{c}) \;\; \forall \hat{c} \in \widehat{\mathcal{C}}\}. \tag{1.5}$$

Theorem 1.5 (Validity of the admissible geometries). Let $\widehat{\mathcal{C}}$ and n be a matched alphabet and round count (Definition 1.3), and let (tag, data) be an admissible geometry with denotation Γ.

(i) If the validation predicate holds, Γ is a non-empty, closed, convex and compact subset of $\Delta(\widehat{\mathcal{C}})$. In particular Definition "Admissible acceptance set" (meat-core/def:admissible-acceptance-set), that $\mathcal{S}_{\mathrm{acc}}$ is a closed convex subset of the probability simplex on $\widehat{\mathcal{C}}$, holds by construction for every admissible geometry whose validation predicate holds.

(ii) For the tag singleton, the validation predicate is necessary and sufficient for Γ to be a non-empty subset of $\Delta(\widehat{\mathcal{C}})$.

(iii) For the tag polytope in canonical box form with $0 \leq \mathbf{q}^- \leq \mathbf{q}^+$, the validation predicate, the condition $\Gamma \neq \varnothing$ and the closed-form condition

$$\sum_{\hat{c} \in \widehat{\mathcal{C}}} q^-(\hat{c}) \;\leq\; 1 \;\leq\; \sum_{\hat{c} \in \widehat{\mathcal{C}}} q^+(\hat{c}) \tag{1.6}$$

are equivalent.

2 Binomial tails

Theorem 2.1 (Binomial tails in closed form)**.** Let $n \geq 1$, $p \in [0,1]$, and let N be a binomial random variable with parameters (n,p), i.e. $\Pr[\mathsf{N} = i] = \binom{n}{i} p^i (1-p)^{n-i}$ for $0 \leq i \leq n$. Write

$$\mathcal{I}_x(a,b) \coloneqq \frac{\int_0^x t^{a-1}(1-t)^{b-1}\,\mathrm{d}t}{\int_0^1 t^{a-1}(1-t)^{b-1}\,\mathrm{d}t}, \qquad x \in [0,1],\ a,b \geq 1. \tag{2.1}$$

Then

$$\Pr[\mathsf{N} \geq k] = \frac{n!}{(k-1)!\,(n-k)!} \int_0^p t^{k-1}(1-t)^{n-k}\,\mathrm{d}t = \mathcal{I}_p(k, n-k+1), \quad 1 \leq k \leq n, \tag{2.2}$$

$$\Pr[\mathsf{N} \leq k] = 1 - \Pr[\mathsf{N} \geq k+1] = \mathcal{I}_{1-p}(n-k, k+1), \qquad 0 \leq k \leq n-1, \tag{2.3}$$

together with the boundary values $\Pr[\mathsf{N} \geq k] = 1$ for $k \leq 0$, $\Pr[\mathsf{N} \geq k] = 0$ for $k > n$, $\Pr[\mathsf{N} \leq k] = 0$ for $k < 0$ and $\Pr[\mathsf{N} \leq k] = 1$ for $k \geq n$. Moreover $p \mapsto \Pr[\mathsf{N} \geq k]$ is non-decreasing on $[0,1]$ for every $k \geq 1$.

Corollary 2.2 (The two strict binomial tails)**.** Let $n \geq 1$, $p \in [0,1]$ and let N be binomial with parameters (n,p), as in Theorem 2.1. Then, for $0 \leq k \leq n$,

$$\begin{aligned} \Pr[\mathsf{N} < k] = \Pr[\mathsf{N} \leq k-1] &= \begin{cases} 0, & k = 0, \\ \mathcal{I}_{1-p}(n-k+1, k), & 1 \leq k \leq n, \end{cases} \\ \Pr[\mathsf{N} > k] = \Pr[\mathsf{N} \geq k+1] &= \begin{cases} \mathcal{I}_p(k+1, n-k), & 0 \leq k \leq n-1, \\ 0, & k = n. \end{cases} \end{aligned} \tag{2.5}$$

Lemma 2.3 (Exact strict tails at a rational success probability)**.** Let $n \geq 1$, let $p \in [0,1] \cap \mathbb{Q}$ and let N be binomial with parameters (n,p), as in Theorem 2.1. For $0 \leq k \leq n$,

$$\Pr[\mathsf{N} < k] = \sum_{i=0}^{k-1} \binom{n}{i} p^i (1-p)^{n-i}, \qquad \Pr[\mathsf{N} > k] = \sum_{i=k+1}^{n} \binom{n}{i} p^i (1-p)^{n-i}, \tag{2.6}$$

an empty sum being 0, and:

(i) both strict tails are rational numbers, obtained from n, k and p by finitely many additions and multiplications of rationals, and they are equal to the closed forms (2.5) of Corollary 2.2;

(ii) for every rational ε, whether $\Pr[\mathsf{N} < k] \leq \varepsilon$ holds, and whether $\Pr[\mathsf{N} > k] \leq \varepsilon$ holds, is decided exactly by evaluating the rational number (2.6) and comparing two rationals, which is a comparison of two integers.

3 The exact box oracle

Theorem 3.1 (Acceptance support and the exact box oracle). Let $\widehat{\mathcal{C}}$ and n be a matched alphabet and round count (Definition 1.3), let $\mathcal{S}_{\mathrm{acc}}$ be a non-empty canonical box, (1.5), with $0 \le \mathbf{q}^- \le \mathbf{q}^+$, and let $g \in \mathbb{Q}^{\widehat{\mathcal{C}}}$.

(i) *(acceptance support)* The set $\widehat{\mathcal{C}}^+ := \{\hat{c} \in \widehat{\mathcal{C}} : q^+(\hat{c}) > 0\}$ contains $\operatorname{supp} \mathbf{q}$ for every $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$; hence it contains the acceptance support $\widehat{\mathcal{C}}_{\mathrm{acc}} = \bigcup_{\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}} \operatorname{supp} \mathbf{q}$ of Definition "Ambient program, pairings, acceptance support and linear minimisation" (meat-core/frank-wolfe/def:first-order-setting), which it need not equal: the box constraints together with the simplex constraint can force $q(\hat{c}) = 0$ at an outcome with $q^+(\hat{c}) > 0$. For a singleton geometry, which declares no box, the union of the supports is $\operatorname{supp} \mathbf{q}^*$, the acceptance support itself.

(ii) *(exact minimum in closed form)* Enumerate $\widehat{\mathcal{C}} = \{c_1, \dots, c_m\}$ so that $g(c_1) \le \cdots \le g(c_m)$. Put $r := 1 - \sum_{\hat{c}} q^-(\hat{c})$ and $D_i := q^+(c_i) - q^-(c_i) \ge 0$, let j^* be the least index with $\sum_{i \le j^*} D_i \ge r$, and define

$$q^{\mathrm{opt}}(c_i) := \begin{cases} q^+(c_i), & i < j^*, \\ q^-(c_i) + \big(r - \sum_{i' < j^*} D_{i'}\big), & i = j^*, \\ q^-(c_i), & i > j^*. \end{cases} \tag{3.1}$$

Then j^* exists, $\mathbf{q}^{\mathrm{opt}} \in \mathcal{S}_{\mathrm{acc}}$, and

$$\langle g, \mathbf{q}^{\mathrm{opt}} \rangle \;=\; \min_{\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}} \langle g, \mathbf{q} \rangle, \tag{3.2}$$

the minimum being attained. All quantities are exact rationals when $g, \mathbf{q}^-, \mathbf{q}^+$ are.

(iii) *(dual witness)* With $B, \mathbf{b}$ as in (1.4), $\lambda := g(c_{j^*})$, $v := -\lambda$ and $\mathbf{u} := (\mathbf{u}^+, \mathbf{u}^-)$ given by $u^+(\hat{c}) := \max\{\lambda - g(\hat{c}), 0\}$ and $u^-(\hat{c}) := \max\{g(\hat{c}) - \lambda, 0\}$, one has $\mathbf{u} \ge 0$,

$$g(\hat{c}) + (B^{\mathsf{T}}\mathbf{u})(\hat{c}) + v = 0 \qquad \text{for every } \hat{c} \in \widehat{\mathcal{C}}, \tag{3.3}$$

in particular the left-hand side is ≥ 0, and

$$-\langle \mathbf{b}, \mathbf{u} \rangle - v \;=\; \langle g, \mathbf{q}^{\mathrm{opt}} \rangle \;=\; \min_{\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}} \langle g, \mathbf{q} \rangle. \tag{3.4}$$

The lower bound $-\langle \mathbf{b}, \mathbf{u} \rangle - v$ on the minimum that this witness gives is therefore *tight* for the canonical box, and every downward-rounded evaluation of $\langle g, \mathbf{q}^{\mathrm{opt}} \rangle$ is a lower bound on $\min_{\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}} \langle g, \mathbf{q} \rangle$.

4 The honest envelope

Definition 4.1 (Honest envelope and its envelope set)**.** An *honest envelope* on $\widehat{\mathcal{C}}$ is a pair $p^{\mathrm{hon}^-}, p^{\mathrm{hon}^+} \in \mathbb{Q}^{\widehat{\mathcal{C}}}$ of exact rational vectors with

$$0 \le p^{\mathrm{hon}^-}(\hat{c}) \le p^{\mathrm{hon}^+}(\hat{c}) \le 1 \quad \forall \hat{c} \in \widehat{\mathcal{C}}, \qquad \sum_{\hat{c}\in\widehat{\mathcal{C}}} p^{\mathrm{hon}^-}(\hat{c}) \le 1 \le \sum_{\hat{c}\in\widehat{\mathcal{C}}} p^{\mathrm{hon}^+}(\hat{c}). \tag{4.1}$$

Its *envelope set* is

$$\mathsf{H} \coloneqq \{\mathbf{p} \in \Delta(\widehat{\mathcal{C}}) : \ p^{\mathrm{hon}^-}(\hat{c}) \le p(\hat{c}) \le p^{\mathrm{hon}^+}(\hat{c}) \ \ \forall \hat{c} \in \widehat{\mathcal{C}}\}, \tag{4.2}$$

the set of honest laws the envelope admits. An honest envelope with $p^{\mathrm{hon}^-} = p^{\mathrm{hon}^+}$ is a *degenerate envelope*; then (4.1) forces the common value to be a law $p^{\mathrm{hon}} \in \Delta(\widehat{\mathcal{C}}) \cap \mathbb{Q}^{\widehat{\mathcal{C}}}$ and $\mathsf{H} = \{p^{\mathrm{hon}}\}$. A single honest law p^{hon} (the *point form*) is read as the degenerate envelope $(p^{\mathrm{hon}}, p^{\mathrm{hon}})$. The pair $(p^{\mathrm{hon}^-}, p^{\mathrm{hon}^+})$ is a canonical box form in the sense of Definition 1.4 (rational, $0 \le p^{\mathrm{hon}^-} \le p^{\mathrm{hon}^+}$), H is its denotation (1.5), and the second condition of (4.1) is its closed-form feasibility condition (1.6).

Lemma 4.2 (The envelope assumption at a degenerate envelope)**.** Let $p^{\mathrm{hon}} \in \Delta(\widehat{\mathcal{C}}) \cap \mathbb{Q}^{\widehat{\mathcal{C}}}$ and let $\widehat{C}_1, \dots, \widehat{C}_n$ be $\widehat{\mathcal{C}}$-valued random variables on one probability space. The following are equivalent:

(a) the statement of Assumption 5.2 holds for the degenerate envelope $(p^{\mathrm{hon}}, p^{\mathrm{hon}})$ of Definition 4.1: there is a filtration $\mathcal{F}_0 \subseteq \cdots \subseteq \mathcal{F}_n$ with $\widehat{C}_j$ $\mathcal{F}_j$-measurable and $\Pr\left[\widehat{C}_j = \hat{c} \,|\, \mathcal{F}_{j-1}\right] = p^{\mathrm{hon}}(\hat{c})$ almost surely, for every j and $\hat{c}$;

(b) the statement of Assumption 5.1 holds: $\widehat{C}_1, \dots, \widehat{C}_n$ are independent and identically distributed with law p^{hon}.

Definition 4.3 (Exact data and conservative evaluation)**.** An evaluation of the acceptance-set construction is a *conservative evaluation* when the honest envelope's vectors $p^{\mathrm{hon}^-}, p^{\mathrm{hon}^+}$ (Definition 4.1; for a single declared law, p^{hon} itself), $\varepsilon_{\mathrm{comp}}$ and the allocation are exact rationals, and every tail comparison an endpoint rule makes (Definition 7.1; for the exact rule, those of Definition 8.1) is decided either exactly or against a bound that rounds the tail *upward*.

5 Assumptions

Assumption 5.1 [**A-HON**] (Honest i.i.d. rounds)**.** (i) On an honest run the acceptance statistics $\widehat{C}_1, \dots, \widehat{C}_n$ are independent and identically distributed with law p^{hon} on $\widehat{\mathcal{C}}$.

Assumption 5.2 [**A-HON-ENV**] (Honest rounds inside a declared envelope)**.** Let $(p^{\mathrm{hon}^-}, p^{\mathrm{hon}^+})$ be the declared honest envelope (Definition 4.1). On an honest run there is a filtration $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \cdots \subseteq \mathcal{F}_n$ on the underlying probability space such that (i) $\widehat{C}_j$ is $\mathcal{F}_j$-measurable for every $j \in \{1, \ldots, n\}$; (ii) for every such j and every $\hat{c} \in \widehat{\mathcal{C}}$, almost surely,

$$p^{\mathrm{hon}^-}(\hat{c}) \;\le\; \Pr\left[\widehat{C}_j = \hat{c} \;\middle|\; \mathcal{F}_{j-1}\right] \;\le\; p^{\mathrm{hon}^+}(\hat{c}). \tag{5.1}$$

Assumption 5.3 [**A-ALPHABET**] (An acceptance value recorded in every round)**.** (i) In every one of the n rounds of the run, an acceptance value in $\widehat{\mathcal{C}}$ is recorded.

Assumption 5.4 [**A-REACH**] (An honest law produced by a feasible state)**.** (i) There exists $\omega^{\mathrm{hon}} \in \mathcal{W}$, the feasible set of the consuming declaration (Definition "Feasible set" (protocol-compiler/def:feasible)), with $\mathbf{\Phi}[\omega^{\mathrm{hon}}] \in \mathsf{H}$, the envelope set of the declared honest envelope (Definition 4.1); for a single declared law p^{hon} (the degenerate envelope, $\mathsf{H} = \{p^{\mathrm{hon}}\}$) this reads $\mathbf{\Phi}[\omega^{\mathrm{hon}}] = p^{\mathrm{hon}}$.

Engine requirements

6 Requirements: the emitted geometry

Requirement 6.1 (Acceptance sets under exactly the two geometry tags). The engine emits every acceptance set under one of the two tags of Definition 1.4 (`geometry_tag`) together with the data of that tag, and a consumer presented with an acceptance set under any other tag refuses it rather than guessing.

Requirement 6.2 (Recorded bounds of a canonical box). The engine emits a geometry in canonical box form with its pair $(\mathbf{q}^-, \mathbf{q}^+)$ recorded under `box_bounds`, beside the rows (1.4) that the pair determines.

Requirement 6.3 (The emitted acceptance support). The engine emits under `acceptance_support` the set $\widehat{\mathcal{C}}^+ = \{\hat{c} \in \widehat{\mathcal{C}} : q^+(\hat{c}) > 0\}$ of Theorem 3.1 (i) for a geometry in canonical box form, and the set $\operatorname{supp} \mathbf{q}^*$ for a singleton geometry. For a box the emitted set is a superset of the acceptance support $\widehat{\mathcal{C}}_{\mathrm{acc}}$ of Definition "Ambient program, pairings, acceptance support and linear minimisation" (meat-core/frank-wolfe/def:first-order-setting) and is never to be substituted for it; for a singleton the emitted set and $\widehat{\mathcal{C}}_{\mathrm{acc}}$ coincide.

Requirement 6.4 (Alphabet and round count taken from the consuming declaration). The engine docks an acceptance set into the declaration a consumer solves only on a matched alphabet and round count (Definition 1.3).

(i) A set it builds, it builds on the acceptance alphabet, in its declared enumeration, of the declaration the consumer solves and on the round count of the one run parameter the consumer reads, and before the set is docked it compares the set's alphabet and enumeration with the declaration's. A disagreement is a hard error of the consumer's input validation: it is a refusal code of neither module, nothing is repaired, and no result is produced.

(ii) A set that the declaration itself carries, with no alphabet of its own, it reads positionally: its k-th coordinate is the k-th acceptance value in the declaration's declared enumeration, and its round count is the one run parameter the consumer reads, as in (i). A set whose number of coordinates differs from the size of the acceptance alphabet is refused as ill formed, and no result is produced.

Binomial box construction

7 Admissible inputs, endpoint rules and the guard

Definition 7.1 (Admissible inputs, endpoint rules and guarded endpoints)**.** Fix $n \geq 1$ and let N be binomial with parameters (n, p), where $p \in [0, 1]$ is rational. An *admissible endpoint rule* R assigns to every $(n, p, \varepsilon^-, \varepsilon^+)$ with rational $\varepsilon^-, \varepsilon^+ \in [0, 1)$ two integers $R^- = R^-(n, p, \varepsilon^-)$ and $R^+ = R^+(n, p, \varepsilon^+)$ such that

(a) *(admissibility)* $R^-, R^+ \in \{0, \ldots, n\}$ and

$$\Pr\left[\mathsf{N} < R^-\right] \leq \varepsilon^-, \qquad \Pr\left[\mathsf{N} > R^+\right] \leq \varepsilon^+; \tag{7.1}$$

(b) *(zero budget)* if $0 < p < 1$, then $R^- = 0$ at $\varepsilon^- = 0$ and $R^+ = n$ at $\varepsilon^+ = 0$;

(c) *(degenerate mass)* $R^- = R^+ = 0$ at $p = 0$ and $R^- = R^+ = n$ at $p = 1$, for every $\varepsilon^{\pm}$.

The rule is *monotone in the allocation* when, the other arguments fixed, $\varepsilon^- \mapsto R^-$ is non-decreasing and $\varepsilon^+ \mapsto R^+$ is non-increasing.

An *admissible input* consists of

(i) a finite, non-empty acceptance alphabet $\widehat{\mathcal{C}}$ with a fixed enumeration;

(ii) a round count, an integer $n \geq 1$;

(iii) an honest input on $\widehat{\mathcal{C}}$: exactly one of a law $p^{\mathrm{hon}} \in \Delta(\widehat{\mathcal{C}})$, the point form, and an honest envelope $(p^{\mathrm{hon}^-}, p^{\mathrm{hon}^+})$ (Definition 4.1); a law in the point form is read as the degenerate envelope $(p^{\mathrm{hon}}, p^{\mathrm{hon}})$;

(iv) a completeness budget $\varepsilon_{\mathrm{comp}} \in [0, 1)$;

(v) a set $\widehat{\mathcal{C}}_{\mathrm{con}} \subseteq \widehat{\mathcal{C}}$ of constrained outcomes;

(vi) an allocation $\varepsilon^-, \varepsilon^+ : \widehat{\mathcal{C}}_{\mathrm{con}} \to \mathbb{Q}_{\geq 0}$ with $\sum_{\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}} (\varepsilon^-(\hat{c}) + \varepsilon^+(\hat{c})) \leq \varepsilon_{\mathrm{comp}}$;

(vii) exactness: every coordinate of the honest input, $\varepsilon_{\mathrm{comp}}$ and every value of the allocation is an exact rational;

(viii) a choice of rule, the exact binomial quantile rule of Definition 8.1 or the Bernstein rule of Proposition 14.2, and a choice of one of two evaluations, by the closed forms or by the enclosures of Proposition 15.3, which on the exact rule decides its tail comparisons: by the enclosures, it gives the rule R_{encl} of that proposition in place of the exact rule;

(ix) consistency with the consumer: $\widehat{\mathcal{C}}$, its enumeration and n are those of the declaration the consumer solves, a matched alphabet and round count (Definition 1.3).

The clauses fix the form of the data — an enumerated set $\widehat{\mathcal{C}}$, an integer n, one honest input that is a vector (the point form) or a pair of vectors (the envelope form), a number $\varepsilon_{\mathrm{comp}}$, a set $\widehat{\mathcal{C}}_{\mathrm{con}}$ with a pair of vectors indexed by it, and a named rule and evaluation — and conditions on data of that form.

Fix an admissible input, with its honest envelope $(p^{\mathrm{hon}^-}, p^{\mathrm{hon}^+})$. For $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}$ write $p^-_{\hat{c}} := p^{\mathrm{hon}^-}(\hat{c})$ and $p^+_{\hat{c}} := p^{\mathrm{hon}^+}(\hat{c})$, let $\mathsf{N}^-_{\hat{c}}$ and $\mathsf{N}^+_{\hat{c}}$ be binomial with parameters $(n, p^-_{\hat{c}})$ and $(n, p^+_{\hat{c}})$ respectively, and abbreviate $R^- := R^-(n, p^-_{\hat{c}}, \varepsilon^-(\hat{c}))$ and $R^+ := R^+(n, p^+_{\hat{c}}, \varepsilon^+(\hat{c}))$: the lower endpoint is the rule's at the envelope's lower value, the upper endpoint the rule's at its upper value. The *guarded endpoints* of R at $\hat{c}$, after the *honest-containment guard*, are

$$k^-(\hat{c}) := \min\{R^-,\ \lfloor n\, p^-_{\hat{c}} \rfloor\}, \qquad k^+(\hat{c}) := \max\{R^+,\ \lceil n\, p^+_{\hat{c}} \rceil\}. \tag{7.2}$$

With the degenerate envelope, $p^-_{\hat{c}} = p^+_{\hat{c}} = p^{\mathrm{hon}}(\hat{c}) =: p_{\hat{c}}$, $\mathsf{N}^-_{\hat{c}}$ and $\mathsf{N}^+_{\hat{c}}$ have the one law of $\mathsf{N}_{\hat{c}}$, binomial $(n, p_{\hat{c}})$, and (7.2) reads $k^-(\hat{c}) = \min\{R^-, \lfloor np_{\hat{c}} \rfloor\}$, $k^+(\hat{c}) = \max\{R^+, \lceil np_{\hat{c}} \rceil\}$ with both rule endpoints at $p_{\hat{c}}$.

The *side-wise reading* is the following. By (7.2), $k^-(\hat{c})$ depends on the envelope only through $p^-_{\hat{c}}$ and $k^+(\hat{c})$ only through $p^+_{\hat{c}}$. A claim of this submodule that fixes one value $p_{\hat{c}}$ and one count $\mathsf{N}_{\hat{c}}$ binomial $(n, p_{\hat{c}})$ and speaks of the guarded endpoints — Definition 8.1 and Propositions 8.2, 14.2 and 15.3 do — states each assertion about the lower endpoint (one naming

k^-, R^-, ε^- and the lower tail) with $p_{\hat{c}} \coloneqq p^-_{\hat{c}}$ and $\mathsf{N}_{\hat{c}} \coloneqq \mathsf{N}^-_{\hat{c}}$, and each assertion about the upper endpoint with $p_{\hat{c}} \coloneqq p^+_{\hat{c}}$ and $\mathsf{N}_{\hat{c}} \coloneqq \mathsf{N}^+_{\hat{c}}$; with the degenerate envelope the two readings coincide and are the claim as written.

Lemma 7.2 (The honest-containment guard for every rule)**.** Let R be an admissible endpoint rule (Definition 7.1) and $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}$; write $p^{\mp} \coloneqq p^{\mp}_{\hat{c}}$ for the envelope's values at $\hat{c}$, $R^- \coloneqq R^-(n, p^-, \varepsilon^-(\hat{c}))$, $R^+ \coloneqq R^+(n, p^+, \varepsilon^+(\hat{c}))$, and let $k^{\mp}(\hat{c})$ be the guarded endpoints (7.2). Then:

(i) *(ordering)*

$$0 \;\le\; k^-(\hat{c}) \;\le\; \lfloor np^- \rfloor \;\le\; \lceil np^+ \rceil \;\le\; k^+(\hat{c}) \;\le\; n; \tag{7.3}$$

(ii) *(the budgets are met)* the maps $k \mapsto \Pr\big[\mathsf{N}^-_{\hat{c}} < k\big]$ and $k \mapsto \Pr\big[\mathsf{N}^+_{\hat{c}} > k\big]$ are non-decreasing and non-increasing respectively on the integers; consequently

$$\Pr\big[\mathsf{N}^-_{\hat{c}} < k^-(\hat{c})\big] \le \varepsilon^-(\hat{c}), \qquad \Pr\big[\mathsf{N}^+_{\hat{c}} > k^+(\hat{c})\big] \le \varepsilon^+(\hat{c}), \tag{7.4}$$

and more generally *every* pair $(\tilde{k}^-, \tilde{k}^+)$ with $0 \le \tilde{k}^- \le R^-$ and $R^+ \le \tilde{k}^+ \le n$ satisfies (7.4) in place of $k^{\mp}(\hat{c})$: enlarging the rule's interval is always admissible;

(iii) *(exact cost of the guard)* relative to the rule's own interval $[R^-, R^+]$, the guard (7.2) enlarges the interval by exactly

$$R^- - k^-(\hat{c}) = \max\big\{0,\; R^- - \lfloor np^- \rfloor\big\}, \qquad k^+(\hat{c}) - R^+ = \max\big\{0,\; \lceil np^+ \rceil - R^+\big\} \tag{7.5}$$

counts on the lower and the upper side respectively;

(iv) *(distance from the honest value)* measured from the envelope's value on each side, the guard costs at most $1/n$ *unconditionally*:

$$0 \le p^- - \frac{k^-(\hat{c})}{n} \le \max\Big\{p^- - \frac{R^-}{n},\; \frac{1}{n}\Big\}, \qquad 0 \le \frac{k^+(\hat{c})}{n} - p^+ \le \max\Big\{\frac{R^+}{n} - p^+,\; \frac{1}{n}\Big\}, \tag{7.6}$$

so on whichever side the guard is active, the emitted endpoint lies within $1/n$ of that side's envelope value.

With the degenerate envelope, $p^- = p^+ = p_{\hat{c}}$, both $\mathsf{N}^{\mp}_{\hat{c}}$ are binomial $(n, p_{\hat{c}})$, and (i)–(iv) are statements about that single law.

8 The exact rule

Definition 8.1 (The exact binomial quantile rule)**.** Fix an admissible input (Definition 7.1), with its round count $n \ge 1$, its honest envelope $(p^{\mathrm{hon}^-}, p^{\mathrm{hon}^+})$

(Definition 4.1; a single law p^{hon} is the degenerate envelope), its constrained outcomes $\widehat{\mathcal{C}}_{\mathrm{con}} \subseteq \widehat{\mathcal{C}}$, its completeness budget $\varepsilon_{\mathrm{comp}} \in [0,1)$ and its budget allocation $\varepsilon^-, \varepsilon^+ : \widehat{\mathcal{C}}_{\mathrm{con}} \to \mathbb{Q}_{\geq 0}$ with $\sum_{\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}} (\varepsilon^-(\hat{c}) + \varepsilon^+(\hat{c})) \leq \varepsilon_{\mathrm{comp}}$; in particular $0 \leq \varepsilon^{\pm}(\hat{c}) \leq \varepsilon_{\mathrm{comp}} < 1$ for every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}$. For $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}$ let $p_{\hat{c}}$ be the rule's argument — by the side-wise reading of Definition 7.1, $p_{\hat{c}} = p^-_{\hat{c}}$ in (8.1) and $p_{\hat{c}} = p^+_{\hat{c}}$ in (8.2), both equal to $p^{\mathrm{hon}}(\hat{c})$ for the degenerate envelope — and let $\mathsf{N}_{\hat{c}}$ be binomial with parameters $(n, p_{\hat{c}})$. Define the *extremal endpoints*

$$\kappa^-(\hat{c}) \coloneqq \max\big\{k \in \{0,\dots,n\} : \ \Pr[\mathsf{N}_{\hat{c}} < k] \leq \varepsilon^-(\hat{c})\big\}, \tag{8.1}$$
$$\kappa^+(\hat{c}) \coloneqq \min\big\{k \in \{0,\dots,n\} : \ \Pr[\mathsf{N}_{\hat{c}} > k] \leq \varepsilon^+(\hat{c})\big\}. \tag{8.2}$$

The *exact binomial quantile rule* is the endpoint rule $R^{\mp}_{\mathrm{ex}}(n, p_{\hat{c}}, \varepsilon^{\mp}(\hat{c})) \coloneqq \kappa^{\mp}(\hat{c})$. Its guarded endpoints are those of (7.2) with $R^{\mp} = \kappa^{\mp}(\hat{c})$, $\kappa^-(\hat{c})$ taken at $p^-_{\hat{c}}$ and $\kappa^+(\hat{c})$ at $p^+_{\hat{c}}$, i.e.

$$k^-(\hat{c}) = \min\big\{\kappa^-(\hat{c}),\ \lfloor n\, p^-_{\hat{c}} \rfloor\big\}, \qquad k^+(\hat{c}) = \max\big\{\kappa^+(\hat{c}),\ \lceil n\, p^+_{\hat{c}} \rceil\big\}.$$

Proposition 8.2 (Properties of the exact binomial quantile rule)**.** Assume a conservative evaluation (Definition 4.3). With the notation of Definition 8.1, for every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}$:

(i) the two sets in (8.1)–(8.2) are non-empty subsets of the finite set $\{0,\dots,n\}$, so $\kappa^{\pm}(\hat{c})$ exist and are unique;

(ii) *(the exact rule is an admissible endpoint rule)* the pair $R^{\mp}_{\mathrm{ex}} = \kappa^{\mp}(\hat{c})$ satisfies Definition 7.1 (a)–(c): $\kappa^{\pm}(\hat{c}) \in \{0,\dots,n\}$ and

$$\Pr\big[\mathsf{N}_{\hat{c}} < \kappa^-(\hat{c})\big] \leq \varepsilon^-(\hat{c}), \qquad \Pr\big[\mathsf{N}_{\hat{c}} > \kappa^+(\hat{c})\big] \leq \varepsilon^+(\hat{c}); \tag{8.3}$$

if $0 < p_{\hat{c}} < 1$ then $\kappa^-(\hat{c}) = 0$ at $\varepsilon^-(\hat{c}) = 0$ and $\kappa^+(\hat{c}) = n$ at $\varepsilon^+(\hat{c}) = 0$; and $\kappa^-(\hat{c}) = \kappa^+(\hat{c}) = 0$ at $p_{\hat{c}} = 0$, $\kappa^-(\hat{c}) = \kappa^+(\hat{c}) = n$ at $p_{\hat{c}} = 1$. Moreover the rule is monotone in the allocation: $\kappa^-(\hat{c})$ is non-decreasing in $\varepsilon^-(\hat{c})$ and $\kappa^+(\hat{c})$ is non-increasing in $\varepsilon^+(\hat{c})$. Consequently Lemma 7.2 applies to the exact rule, and its guarded endpoints $k^{\mp}(\hat{c})$ satisfy (7.3) and (7.4);

(iii) *(extremality and admissible relaxation)* $\kappa^-(\hat{c})$ is the largest and $\kappa^+(\hat{c})$ the smallest integer in $\{0,\dots,n\}$ satisfying the respective inequality of (8.3), and *every* pair $(\tilde{k}^-, \tilde{k}^+)$ with $0 \leq \tilde{k}^- \leq \kappa^-(\hat{c})$ and $\kappa^+(\hat{c}) \leq \tilde{k}^+ \leq n$ satisfies (7.4) in place of $k^{\mp}(\hat{c})$. Enlarging the interval is therefore always admissible; shrinking it beyond $[\kappa^-, \kappa^+]$ is not;

(iv) the defining inequalities are the closed forms (2.5) of Corollary 2.2, so $\kappa^{\pm}(\hat{c})$ are computable from n, $p_{\hat{c}}$ and the allocation alone (Lemma 2.3);

(v) *(degenerate honest mass)* if $p_{\hat{c}} = 0$ then $k^-(\hat{c}) = k^+(\hat{c}) = 0$, and if $p_{\hat{c}} = 1$ then $k^-(\hat{c}) = k^+(\hat{c}) = n$; in both cases $\varepsilon^-(\hat{c})$ and $\varepsilon^+(\hat{c})$ are unspent, both tails in (7.4) being exactly zero;

(vi) *(exact cost of the guard for this rule)* write $p = p_{\hat{c}}$. Relative to the raw quantile interval $[\kappa^-(\hat{c}), \kappa^+(\hat{c})]$, the guard (7.2) enlarges the interval by exactly (7.5) with $R^{\mp} = \kappa^{\mp}(\hat{c})$, and neither enlargement is bounded by one count: for every $n \geq 1$ and every $p \in (0,1)$, an allocation with $\varepsilon^-(\hat{c}) \geq 1 - p^n$ — admissible whenever $\varepsilon_{\text{comp}}$ permits, since $1 - p^n < 1$ — gives $\kappa^-(\hat{c}) = n$ and a lower enlargement of $n - \lfloor np \rfloor$ counts, that is, at least $1 - p$ in frequency *uniformly in* n; symmetrically $\varepsilon^+(\hat{c}) \geq 1 - (1-p)^n$ gives $\kappa^+(\hat{c}) = 0$ and an upper enlargement of $\lceil np \rceil$ counts. The exact conditions are

$$\kappa^-(\hat{c}) - k^-(\hat{c}) = 0 \iff \varepsilon^-(\hat{c}) < \Pr\left[\mathsf{N}_{\hat{c}} \leq \lfloor np \rfloor\right], \tag{8.4}$$
$$\kappa^-(\hat{c}) - k^-(\hat{c}) \leq 1 \iff \varepsilon^-(\hat{c}) < \Pr\left[\mathsf{N}_{\hat{c}} \leq \lfloor np \rfloor + 1\right], \tag{8.5}$$
$$k^+(\hat{c}) - \kappa^+(\hat{c}) = 0 \iff \varepsilon^+(\hat{c}) < \Pr\left[\mathsf{N}_{\hat{c}} \geq \lceil np \rceil\right], \tag{8.6}$$
$$k^+(\hat{c}) - \kappa^+(\hat{c}) \leq 1 \iff \varepsilon^+(\hat{c}) < \Pr\left[\mathsf{N}_{\hat{c}} \geq \lceil np \rceil - 1\right]. \tag{8.7}$$

Measured instead from the honest value, the guard costs at most $1/n$ unconditionally, (7.6) of Lemma 7.2 (iv);

(vii) *(evaluation against upper bounds)* let U_k^- and U_k^+, for $k \in \{0, \ldots, n\}$, be rationals determined by n, $p_{\hat{c}}$ and k alone, with $U_k^- \geq \Pr[\mathsf{N}_{\hat{c}} < k]$, $U_k^+ \geq \Pr[\mathsf{N}_{\hat{c}} > k]$, $U_0^- = 0$ and $U_n^+ = 0$, and with every $U_k^{\mp}$ equal to its tail when $p_{\hat{c}} \in \{0, 1\}$. Put

$$\tilde{\kappa}^-(\hat{c}) \coloneqq \max\left\{k \in \{0, \ldots, n\} : U_j^- \leq \varepsilon^-(\hat{c}) \text{ for } 1 \leq j \leq k\right\}, \qquad \tilde{\kappa}^+(\hat{c}) \coloneqq \min\left\{k \in \{0, \ldots, n\} : \right.$$

Then $\tilde{R}^{\mp} = \tilde{\kappa}^{\mp}(\hat{c})$ is an admissible endpoint rule (Definition 7.1 (a)–(c)), monotone in the allocation, with $\tilde{\kappa}^-(\hat{c}) \leq \kappa^-(\hat{c})$ and $\tilde{\kappa}^+(\hat{c}) \geq \kappa^+(\hat{c})$, and it is the exact rule when every $U_k^{\mp}$ equals its tail. Consequently Lemma 7.2 applies to it.

9 The construction map

Definition 9.1 (The construction map)**.** Let $(\widehat{\mathcal{C}}, p^{\text{hon}-}, p^{\text{hon}+}, n, \varepsilon_{\text{comp}}, \widehat{\mathcal{C}}_{\text{con}}, \varepsilon^-, \varepsilon^+)$ be the data of an admissible input (Definition 7.1), where $(p^{\text{hon}-}, p^{\text{hon}+})$ is its honest envelope (Definition 4.1) — an input p^{hon} in the point form being read as the degenerate envelope $(p^{\text{hon}}, p^{\text{hon}})$ — and let R be an admissible endpoint rule (Definition 7.1). With the guarded endpoints $k^{\pm}$ of R, (7.2), define

$$q^-(\hat{c}) \coloneqq \begin{cases} k^-(\hat{c})/n, & \hat{c} \in \widehat{\mathcal{C}}_{\text{con}}, \\ 0, & \hat{c} \notin \widehat{\mathcal{C}}_{\text{con}}, \end{cases} \qquad q^+(\hat{c}) \coloneqq \begin{cases} k^+(\hat{c})/n, & \hat{c} \in \widehat{\mathcal{C}}_{\text{con}}, \\ 1, & \hat{c} \notin \widehat{\mathcal{C}}_{\text{con}}. \end{cases} \tag{9.1}$$

The *construction map* Acc_R — written Acc where the rule is clear or is the default — returns the polytope geometry in canonical box form with these $(\mathbf{q}^-, \mathbf{q}^+)$, i.e. the data (1.4), and

$$\mathcal{S}_{\mathrm{acc}} \;\coloneqq\; \{\mathbf{q} \in \Delta(\widehat{\mathcal{C}}) :\; q^-(\hat{c}) \le q(\hat{c}) \le q^+(\hat{c}) \;\; \forall \hat{c} \in \widehat{\mathcal{C}}\}. \tag{9.2}$$

Theorem 9.2 (Properties of the constructed box). On every admissible input (Definition 7.1) and for every admissible endpoint rule R, under a conservative evaluation (Definition 4.3):

(i) Acc_R is well defined, and all of its output data are exact rationals: each $q^{\pm}(\hat{c})$ is either an integer multiple of $1/n$ or one of $0, 1$, and the entries of B lie in $\{0, \pm 1\}$;

(ii) $0 \le \mathbf{q}^- \le \mathbf{q}^+ \le \mathbf{1}$ componentwise, so the pair $(\mathbf{q}^-, \mathbf{q}^+)$ meets the conditions of the canonical box form of Definition 1.4;

(iii) $\mathsf{H} \subseteq \mathcal{S}_{\mathrm{acc}}$, where H is the envelope set (4.2) of the input envelope: every honest law the envelope admits is accepted, and for the degenerate envelope this reads $p^{\mathrm{hon}} \in \mathcal{S}_{\mathrm{acc}}$. The envelope set is non-empty; consequently $\mathcal{S}_{\mathrm{acc}} \neq \varnothing$ and, by Theorem 1.5 (iii), $\sum_{\hat{c}} q^-(\hat{c}) \le 1 \le \sum_{\hat{c}} q^+(\hat{c})$;

(iv) $\mathcal{S}_{\mathrm{acc}}$ is a non-empty, closed, convex, compact subset of $\Delta(\widehat{\mathcal{C}})$; Definition "Admissible acceptance set" (meat-core/def:admissible-acceptance-set) holds for the output of Acc_R by construction, for every admissible input and every admissible rule.

10 Completeness

Lemma 10.1 (Binomial dominance under bounded conditional success probabilities)**.** Let $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \cdots \subseteq \mathcal{F}_n$ be a filtration on a probability space, $n \ge 1$, and let $\chi_1, \dots, \chi_n$ be $\{0,1\}$-valued random variables with χ_j $\mathcal{F}_j$-measurable. Let $0 \le a \le b \le 1$ and suppose that, for every $j \in \{1, \dots, n\}$, almost surely

$$a \;\le\; P_j \;\coloneqq\; \Pr\left[\chi_j = 1 \;\middle|\; \mathcal{F}_{j-1}\right] \;\le\; b. \tag{10.1}$$

Put $S \coloneqq \sum_{j=1}^{n} \chi_j$, and for $m \in \{0, \dots, n\}$, $r \in [0,1]$ and an integer k write $G_m^r(k) \coloneqq \Pr[\mathrm{B}_m^r \ge k]$, where B_m^r is binomial with parameters (m, r) ($\mathrm{B}_0^r = 0$). Then for every integer k

$$\Pr[S \ge k] \;\le\; \Pr\left[\mathrm{B}_n^b \ge k\right], \qquad \Pr[S \le k] \;\le\; \Pr\left[\mathrm{B}_n^a \le k\right], \tag{10.2}$$

and equivalently

$$\Pr[S > k] \;\le\; \Pr\left[\mathrm{B}_n^b > k\right], \qquad \Pr[S < k] \;\le\; \Pr\left[\mathrm{B}_n^a < k\right]. \tag{10.3}$$

Theorem 10.2 (The completeness bound). Assume a conservative evaluation (Definition 4.3), and assume A-HON-ENV (Assumption 5.2) for the honest envelope $(p^{\mathrm{hon}^-}, p^{\mathrm{hon}^+})$ of the input: on an honest run there is a filtration $\mathcal{F}_0 \subseteq \cdots \subseteq \mathcal{F}_n$ with $\widehat{C}_j$ $\mathcal{F}_j$-measurable and $p^{\mathrm{hon}^-}(\hat{c}) \le \Pr\left[\widehat{C}_j = \hat{c} \,|\, \mathcal{F}_{j-1}\right] \le p^{\mathrm{hon}^+}(\hat{c})$ almost surely, for every j and $\hat{c}$. Let $\mathcal{S}_{\mathrm{acc}}$ be the output of Acc_R (Definition 9.1) for an admissible endpoint rule R (Definition 7.1). Then

$$\Pr\left[\,\mathrm{freq}_{\widehat{C}_1^n} \notin \mathcal{S}_{\mathrm{acc}}\right] \le \varepsilon^\star := \sum_{\hat{c}\in\widehat{\mathcal{C}}_{\mathrm{con}}} \Big(\Pr\left[\mathsf{N}_{\hat{c}}^- < k^-(\hat{c})\right] + \Pr\left[\mathsf{N}_{\hat{c}}^+ > k^+(\hat{c})\right]\Big) \le \varepsilon_{\mathrm{comp}}, \tag{10.6}$$

where $\mathsf{N}_{\hat{c}}^-$ and $\mathsf{N}_{\hat{c}}^+$ are binomial with parameters $(n, p^{\mathrm{hon}^-}(\hat{c}))$ and $(n, p^{\mathrm{hon}^+}(\hat{c}))$ respectively (Definition 7.1). Equivalently, the honest run passes the acceptance test, the event Ω_{AT} of Definition "Fixed-length post-processing model" (meat-core/def:postprocessing-model), with probability $\Pr[\Omega_{\mathrm{AT}}] = \Pr\left[\mathrm{freq}_{\widehat{C}_1^n} \in \mathcal{S}_{\mathrm{acc}}\right] \ge 1 - \varepsilon^\star \ge 1 - \varepsilon_{\mathrm{comp}}$. For the degenerate envelope both $\mathsf{N}_{\hat{c}}^{\mp}$ are binomial $(n, p^{\mathrm{hon}}(\hat{c}))$.

11 Honest containment and infeasibility

Theorem 11.1 (Honest containment and infeasibility)**.** Let $\mathcal{S}_{\mathrm{acc}}$ be the output of Acc_R on an admissible input (Definition 7.1), for any admissible endpoint rule R, under a conservative evaluation (Definition 4.3).

(i) $\mathsf{H} \subseteq \mathcal{S}_{\mathrm{acc}}$, where H is the envelope set (4.2) of the input envelope, for every admissible $\varepsilon_{\mathrm{comp}} \in [0, 1)$, every $\widehat{\mathcal{C}}_{\mathrm{con}}$ and every admissible allocation; for the degenerate envelope, $p^{\mathrm{hon}} \in \mathcal{S}_{\mathrm{acc}}$.

(ii) Assume A-REACH (Assumption 5.4): there exists $\omega^{\mathrm{hon}} \in \mathcal{W}$, the feasible set of the consuming declaration $\mathfrak{D}$ (Definition "Feasible set" (protocol-compiler/def:feasible)), with $\mathbf{p}^{\mathrm{hon}} := \mathbf{\Phi}[\omega^{\mathrm{hon}}] \in \mathsf{H}$ (for the degenerate envelope, $\mathbf{\Phi}[\omega^{\mathrm{hon}}] = p^{\mathrm{hon}}$). Then the pair $(\mathbf{p}^{\mathrm{hon}}, \omega^{\mathrm{hon}})$ satisfies $\mathbf{p}^{\mathrm{hon}} \in \mathcal{S}_{\mathrm{acc}}$, $\omega^{\mathrm{hon}} \in \mathcal{W}$ and $\mathbf{p}^{\mathrm{hon}} \ll \mathbf{\Phi}[\omega^{\mathrm{hon}}]$. Consequently the condition

no $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$ is absolutely continuous with respect to $\mathbf{\Phi}[\omega]$ for any $\omega \in \mathcal{W}$ (11.1)

is *false* for this $\mathcal{S}_{\mathrm{acc}}$.

(iii) *(refusals come from the input)* The validation predicate of Theorem 1.5 (iii) is decidable in closed form, and on every admissible input it is satisfied by the output of Acc_R unconditionally, for every admissible rule. Hence the construction never fails on an admissible input, and a refusal can arise only from an input that is not admissible. On data of the form

of an admissible input (Definition 7.1), the violations of its conditions that are decidable from this module's own inputs are exactly:

(1) $|\widehat{\mathcal{C}}| = 0$;

(2) $|\widehat{\mathcal{C}}|$ not finite;

(3) $n < 1$;

(4) an honest input in the point form with $p^{\mathrm{hon}} \notin \Delta(\widehat{\mathcal{C}})$, including a coordinate outside $\widehat{\mathcal{C}}$;

(5) an honest input in the envelope form violating (4.1): some entry of p^{hon^-} or p^{hon^+} outside $[0,1]$, some $p^{\mathrm{hon}^-}(\hat{c}) > p^{\mathrm{hon}^+}(\hat{c})$, $\sum_{\hat{c}} p^{\mathrm{hon}^-}(\hat{c}) > 1$, $\sum_{\hat{c}} p^{\mathrm{hon}^+}(\hat{c}) < 1$, or a coordinate outside $\widehat{\mathcal{C}}$;

(6) $\varepsilon_{\mathrm{comp}} \notin [0,1)$;

(7) some $\varepsilon^{\pm}(\hat{c}) < 0$;

(8) $\sum_{\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}} (\varepsilon^-(\hat{c}) + \varepsilon^+(\hat{c})) > \varepsilon_{\mathrm{comp}}$;

(9) $\widehat{\mathcal{C}}_{\mathrm{con}} \not\subseteq \widehat{\mathcal{C}}$;

(10) some coordinate of p^{hon} or of $p^{\mathrm{hon}^{\mp}}$, or $\varepsilon_{\mathrm{comp}}$, or some $\varepsilon^{\pm}(\hat{c})$ is not an exact rational, violating Definition 4.3;

(11) the requested endpoint rule, or the requested evaluation, is none of those of clause (viii) of Definition 7.1.

Exactly one clause of an admissible input is *not* decidable from this module's inputs: clause (ix) of Definition 7.1, the condition of Definition 1.3 that $\widehat{\mathcal{C}}$, its enumeration and n are those of the same declaration $\mathfrak{D}$ the consumer solves. That clause is a relation between artefacts, not a property of an input.

12 Monotonicity and limiting cases

Definition 12.1 (Monotone allocation rule)**.** A *monotone allocation rule* assigns to each budget $e \in [0,1)$ an allocation $\varepsilon^{\pm}_{(e)}$ on $\widehat{\mathcal{C}}_{\mathrm{con}}$ such that $\sum_{\hat{c}} (\varepsilon^-_{(e)}(\hat{c}) + \varepsilon^+_{(e)}(\hat{c})) \le e$ and each $e \mapsto \varepsilon^{\pm}_{(e)}(\hat{c})$ is non-decreasing. The *uniform rule* $\varepsilon^{\pm}_{(e)}(\hat{c}) = e/(2|\widehat{\mathcal{C}}_{\mathrm{con}}|)$ is monotone.

Lemma 12.2 (Widening the box keeps its guarantees)**.** Let $\mathcal{S}_{\mathrm{acc}}$ be the output of Acc_R (Definition 9.1) on an admissible input (Definition 7.1) for an admissible endpoint rule R, with box bounds $\mathbf{q}^-, \mathbf{q}^+$ (9.1) and guarded endpoints $k^{\mp}$ (7.2). Let $\tilde{\mathbf{q}}^-, \tilde{\mathbf{q}}^+ \in \mathbb{R}^{\widehat{\mathcal{C}}}$ satisfy $\tilde{q}^-(\hat{c}) \le q^-(\hat{c})$ and $\tilde{q}^+(\hat{c}) \ge q^+(\hat{c})$ for every $\hat{c} \in \widehat{\mathcal{C}}$, and put

$$\mathcal{S}'_{\mathrm{acc}} \;\coloneqq\; \{\mathbf{q} \in \Delta(\widehat{\mathcal{C}}) :\; \tilde{q}^-(\hat{c}) \le q(\hat{c}) \le \tilde{q}^+(\hat{c}) \;\; \forall \hat{c} \in \widehat{\mathcal{C}}\}. \tag{12.1}$$

(i) *(containment)* $\mathcal{S}_{\mathrm{acc}} \subseteq \mathcal{S}'_{\mathrm{acc}}$.

(ii) *(failure probability)* For all $\widehat{\mathcal{C}}$-valued random variables $\widehat{C}_1, \dots, \widehat{C}_n$ on one probability space,

$$\Pr\left[\mathrm{freq}_{\widehat{C}_1^n} \notin \mathcal{S}'_{\mathrm{acc}}\right] \;\le\; \Pr\left[\mathrm{freq}_{\widehat{C}_1^n} \notin \mathcal{S}_{\mathrm{acc}}\right].$$

(iii) *(infima)* For every function $\Psi : \Delta(\widehat{\mathcal{C}}) \to [-\infty, +\infty]$,

$$\inf_{\mathbf{q} \in \mathcal{S}'_{\mathrm{acc}}} \Psi(\mathbf{q}) \;\le\; \inf_{\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}} \Psi(\mathbf{q}). \tag{12.2}$$

(iv) *(integer endpoints)* Let $\tilde{k}^-, \tilde{k}^+ : \widehat{\mathcal{C}}_{\mathrm{con}} \to \mathbb{Z}$ satisfy $0 \le \tilde{k}^-(\hat{c}) \le k^-(\hat{c})$ and $k^+(\hat{c}) \le \tilde{k}^+(\hat{c}) \le n$ for every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}$, and let $\tilde{\mathbf{q}}^{\mp}$ be given by (9.1) with $\tilde{k}^{\mp}$ in place of $k^{\mp}$. Then $\tilde{\mathbf{q}}^{\mp}$ satisfy the hypothesis above, so (i)–(iii) hold; the pair $(\tilde{\mathbf{q}}^-, \tilde{\mathbf{q}}^+)$ meets the conditions of the canonical box form of Definition 1.4, and its denotation (1.5) is $\mathcal{S}'_{\mathrm{acc}}$; and for every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}$, with $\mathsf{N}_{\hat{c}}^{\mp}$ as in Definition 7.1,

$$\Pr\left[\mathsf{N}_{\hat{c}}^- < \tilde{k}^-(\hat{c})\right] \le \Pr\left[\mathsf{N}_{\hat{c}}^- < k^-(\hat{c})\right] \le \varepsilon^-(\hat{c}), \qquad \Pr\left[\mathsf{N}_{\hat{c}}^+ > \tilde{k}^+(\hat{c})\right] \le \Pr\left[\mathsf{N}_{\hat{c}}^+ > k^+(\hat{c})\right] \le \varepsilon^+(\hat{c}). \tag{12.3}$$

Summed over $\widehat{\mathcal{C}}_{\mathrm{con}}$, the tails at $\tilde{k}^{\mp}$ therefore total at most the tails at $k^{\mp}$, and at most $\varepsilon_{\mathrm{comp}}$.

(v) *(not the reverse)* Replacing $k^-(\hat{c})$ by a larger integer or $k^+(\hat{c})$ by a smaller one keeps, in general, neither the budgets of (12.3) nor the tail sum at $k^{\mp}$ as a bound on the probability that the frequency vector leaves the set. Take $\widehat{\mathcal{C}} = \{\hat{c}_0, \hat{c}_1\}$, the degenerate envelope at $p^{\mathrm{hon}} = (\frac{1}{2}, \frac{1}{2})$ (Definition 4.1), $n = 3$, $\widehat{\mathcal{C}}_{\mathrm{con}} = \{\hat{c}_0\}$, $\varepsilon^-(\hat{c}_0) = \varepsilon^+(\hat{c}_0) = \frac{2}{5}$, $\varepsilon_{\mathrm{comp}} = \frac{4}{5}$ and the exact binomial quantile rule of Definition 8.1. Then $k^-(\hat{c}_0) = 1$, $k^+(\hat{c}_0) = 2$, and the tails at these endpoints sum to $\frac{1}{4}$. Replacing $k^-(\hat{c}_0)$ by 2, or $k^+(\hat{c}_0)$ by 1, gives a tail of $\frac{1}{2} > \frac{2}{5}$ on the replaced side; and for $\widehat{C}_1, \widehat{C}_2, \widehat{C}_3$ independent with law p^{hon}, the frequency vector lies outside the set (9.2) built from the replaced endpoints with probability $\frac{5}{8} > \frac{1}{4}$, while it lies outside $\mathcal{S}_{\mathrm{acc}}$ with probability $\frac{1}{4}$.

Theorem 12.4 (Monotonicity in the budget and limiting cases)**.** Fix $\widehat{\mathcal{C}}$, an honest envelope $(p^{\mathrm{hon}^-}, p^{\mathrm{hon}^+})$ with envelope set H (Definition 4.1; a single law p^{hon} is the degenerate envelope, $\mathsf{H} = \{p^{\mathrm{hon}}\}$), n, $\widehat{\mathcal{C}}_{\mathrm{con}}$ and an admissible endpoint rule R (Definition 7.1) under a conservative evaluation (Definition 4.3), write $p_{\hat{c}}^{\mp}$ for the envelope's values as there, and write $\mathcal{S}_{\mathrm{acc}}(e)$ for the output of Acc_R at budget e under a fixed monotone allocation rule (Definition 12.1).

(i) *(budget monotonicity: the set shrinks as the budget grows)* If R is monotone in the allocation and $0 \le e_1 \le e_2 < 1$, then $\mathcal{S}_{\mathrm{acc}}(e_2) \subseteq \mathcal{S}_{\mathrm{acc}}(e_1)$.

(ii) *(consequence for any consumer infimum)* Under the hypothesis of (i), for every function $\Psi : \Delta(\widehat{\mathcal{C}}) \to [-\infty, +\infty]$ and $e_1 \le e_2$,

$$\inf_{\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}(e_1)} \Psi(\mathbf{q}) \;\le\; \inf_{\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}(e_2)} \Psi(\mathbf{q}). \tag{12.4}$$

In particular, taking $\Psi(\mathbf{q}) = \inf_{\omega \in \mathcal{W}} O_\alpha(\mathbf{q}, \omega)$, the quantity $\underline{h}_\alpha$, which Definition "Feasible set and computable objective" (meat-core/def:feasible) defines as $\inf_{\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}} \inf_{\omega \in \mathcal{W}} O_\alpha$, is *non-decreasing* in $\varepsilon_{\mathrm{comp}}$: a weaker completeness guarantee buys a larger or equal entropy bound.

(iii) *(limit $\varepsilon_{\mathrm{comp}} = 0$, every rule)* At $e = 0$ every $\varepsilon^{\pm}(\hat{c}) = 0$ and

$$\mathcal{S}_{\mathrm{acc}}(0) = \big\{ \mathbf{q} \in \Delta(\widehat{\mathcal{C}}) : q(\hat{c}) = 0 \text{ for } \hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}} \text{ with } p^{+}_{\hat{c}} = 0, \\ q(\hat{c}) = 1 \text{ for } \hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}} \text{ with } p^{-}_{\hat{c}} = 1 \big\},$$

with $\varepsilon^\star = 0$. If $p^{+}_{\hat{c}} > 0$ and $p^{-}_{\hat{c}} < 1$ for every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}$ (for the degenerate envelope: if $0 < p^{\mathrm{hon}}(\hat{c}) < 1$ for every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}$), then $\mathcal{S}_{\mathrm{acc}}(0) = \Delta(\widehat{\mathcal{C}})$: a zero completeness budget buys the whole simplex and no constraint at all.

(iv) *(limit $n \to \infty$, the exact rule)* Let R be the exact rule of Definition 8.1, let $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}$ and write $p^{\mp} = p^{\mp}_{\hat{c}}$. Then

$$0 \le q^{+}(\hat{c}) - p^{+} \le \sqrt{\frac{p^{+}(1-p^{+})}{n\,\varepsilon^{+}(\hat{c})}} + \frac{1}{n}, \qquad 0 \le p^{-} - q^{-}(\hat{c}) \le \sqrt{\frac{p^{-}(1-p^{-})}{n\,\varepsilon^{-}(\hat{c})}} + \frac{1}{n}, \tag{12.5}$$

the first whenever $0 < p^{+} < 1$ and $\varepsilon^{+}(\hat{c}) > 0$, the second whenever $0 < p^{-} < 1$ and $\varepsilon^{-}(\hat{c}) > 0$. Consequently, if $\widehat{\mathcal{C}}_{\mathrm{con}} = \widehat{\mathcal{C}}$, $0 < p^{\mp}_{\hat{c}} < 1$ and $\varepsilon^{\pm}(\hat{c}) > 0$ for all $\hat{c}$, and the allocation is held fixed as n grows (for instance the uniform rule at a fixed $e > 0$), then every $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$ satisfies $p^{-}_{\hat{c}} - O(n^{-1/2}) \le q(\hat{c}) \le p^{+}_{\hat{c}} + O(n^{-1/2})$ uniformly in $\hat{c}$: $\mathcal{S}_{\mathrm{acc}}$ lies coordinatewise within $O(n^{-1/2})$ of the box $\prod_{\hat{c}} [p^{-}_{\hat{c}}, p^{+}_{\hat{c}}]$, and it contains H by Theorem 9.2 (iii). For the degenerate envelope, $\sup_{\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}} \|\mathbf{q} - p^{\mathrm{hon}}\|_\infty = O(n^{-1/2})$ and $\mathcal{S}_{\mathrm{acc}}$ converges to $\{p^{\mathrm{hon}}\}$ in Hausdorff distance.

(v) *(no monotonicity in n, with a counterexample; the exact rule)* For the exact rule, $\mathcal{S}_{\mathrm{acc}}$ is *not* monotone in n, already for a degenerate envelope. Take $\widehat{\mathcal{C}} = \{\hat{c}_0, \hat{c}_1\}$, the degenerate envelope at $p^{\mathrm{hon}} = (\frac{1}{2}, \frac{1}{2})$, $\widehat{\mathcal{C}}_{\mathrm{con}} = \{\hat{c}_0\}$ and $\varepsilon^{-}(\hat{c}_0) = \varepsilon^{+}(\hat{c}_0) = \frac{2}{5}$ (so $\varepsilon_{\mathrm{comp}} = \frac{4}{5}$). Then

$$n = 2: \quad [k^{-}, k^{+}] = [1, 1], \quad \mathcal{S}_{\mathrm{acc}} = \{(\tfrac{1}{2}, \tfrac{1}{2})\}; \\ n = 3: \quad [k^{-}, k^{+}] = [1, 2], \quad \mathcal{S}_{\mathrm{acc}} = \{\mathbf{q} \in \Delta(\widehat{\mathcal{C}}) : \tfrac{1}{3} \le q(\hat{c}_0) \le \tfrac{2}{3}\},$$

so $\mathcal{S}_{\mathrm{acc}}(n{=}2) \subsetneq \mathcal{S}_{\mathrm{acc}}(n{=}3)$: the set is strictly larger at the larger round count.

(vi) *(degenerate alphabet $|\widehat{\mathcal{C}}| = 1$, every rule)* If $\widehat{\mathcal{C}} = \{\hat{c}_0\}$ then, for every $n \geq 1$, every $\varepsilon_{\mathrm{comp}} \in [0, 1)$, every $\widehat{\mathcal{C}}_{\mathrm{con}} \subseteq \widehat{\mathcal{C}}$ and every admissible allocation,

$$p^{\mathrm{hon}^+}(\hat{c}_0) = 1, \qquad \mathsf{H} = \mathcal{S}_{\mathrm{acc}} = \Delta(\widehat{\mathcal{C}}) = \{\delta_{\hat{c}_0}\},$$

and $\varepsilon^\star = 0$ whenever $p^{\mathrm{hon}^-}(\hat{c}_0) = 1$, in particular for the degenerate envelope, which is then $p^{\mathrm{hon}} = \delta_{\hat{c}_0}$. The emitted set is therefore constant in all of the construction's parameters on a one-letter alphabet, being at once the whole simplex and a single point; for the degenerate envelope so are its box data, and completeness is perfect. For a non-degenerate envelope ($p^{\mathrm{hon}^-}(\hat{c}_0) < 1$, which (4.1) permits here) the lower box bound $q^-(\hat{c}_0)$ and the reported $\varepsilon^\star$ may be positive and may vary with the parameters; the reported figure is still a valid bound (by Theorem 10.2), merely a loose one, the true honest failure probability being 0.

13 Elementary enclosures

Lemma 13.1 (Rational bounds on the logarithm from integer square roots)**.** Let $x > 0$ be real and $m \geq 1$ an integer. Then

$$m\big(1 - x^{-1/m}\big) \;\leq\; \ln x \;\leq\; m\big(x^{1/m} - 1\big), \tag{13.1}$$

with equality on either side if and only if $x = 1$, and both slacks are bounded by

$$\max\Big\{m\big(x^{1/m}-1\big)-\ln x,\;\; \ln x - m\big(1-x^{-1/m}\big)\Big\} \;\leq\; \frac{(\ln x)^2}{2m}\, e^{|\ln x|/m}. \tag{13.2}$$

Consequently, if l, u are rationals with $0 < l \leq x^{1/m} \leq u$, then

$$m\Big(1 - \frac{1}{l}\Big) \;\leq\; \ln x \;\leq\; m\,(u - 1), \tag{13.3}$$

and for rational $x = a/b$ (a, b positive integers) and $m = 2^j$, such l, u of any prescribed positive width are computed by j integer square roots: with $\mathrm{isqrt}(N) := \lfloor \sqrt{N} \rfloor$ for an integer $N \geq 0$, one has for every rational $y = c/d > 0$

$$\frac{\mathrm{isqrt}(c\,d\,4^s)}{d\,2^s} \;\leq\; \sqrt{y} \;<\; \frac{\mathrm{isqrt}(c\,d\,4^s) + 1}{d\,2^s}, \tag{13.4}$$

the two bounds differing by $1/(d\,2^s) \leq 2^{-s}$, and iterating (13.4) j times, the lower bound always taken from the previous lower bound and the upper from the previous upper, gives $l \leq x^{1/2^j} \leq u$ whose width $u - l$ is an explicit rational the caller reads off and drives below any prescribed bound by the precisions s of the steps.

14 The Bernstein rule

Proposition 14.2 (The Bernstein endpoint rule)**.** Assume a conservative evaluation (Definition 4.3). Fix an integer $j \geq 1$ and a precision $s \geq 0$ (parameters of the evaluation). The *Bernstein rule* R_{B} assigns to $(n, p, \varepsilon^-, \varepsilon^+)$, with $v := np(1-p)$:

- if $p = 0$: $R_{\mathrm{B}}^- = R_{\mathrm{B}}^+ = 0$; if $p = 1$: $R_{\mathrm{B}}^- = R_{\mathrm{B}}^+ = n$;
- otherwise, for each sign separately: if $\varepsilon^- = 0$ then $R_{\mathrm{B}}^- = 0$, and if $\varepsilon^+ = 0$ then $R_{\mathrm{B}}^+ = n$; if $\varepsilon^{\mp} > 0$, let $L^{\mp}$ be the rational upper bound $m(u-1)$ of (13.3) on $\ln(1/\varepsilon^{\mp})$ with $m = 2^j$ and u the upper root enclosure of (13.4) at precision s, let

$$t^{\mp} \;:=\; \text{a rational} \;\geq\; \frac{L^{\mp}}{3} + \sqrt{\left(\frac{L^{\mp}}{3}\right)^2 + 2vL^{\mp}} \tag{14.4}$$

  obtained by rounding the square root upward with (13.4), and put

$$R_{\mathrm{B}}^- \;:=\; \max\left\{0,\; \lfloor np - t^- \rfloor\right\}, \qquad R_{\mathrm{B}}^+ \;:=\; \min\left\{n,\; \lceil np + t^+ \rceil\right\}. \tag{14.5}$$

Then:

(i) R_{B} is an admissible endpoint rule in the sense of Definition 7.1 (a)–(c), computed from n, p and the allocation by finitely many operations on integers and rationals (integer square roots included) and no others; Lemma 7.2 therefore applies to it;

(ii) R_{B} is monotone in the allocation, provided every rounding in $L^{\mp}$ and $t^{\mp}$ is a non-decreasing function of its real argument. The integer-square-root rounding of (13.4) is *not* such a function of a rational argument as written (its excess over the root depends on the denominator), but it becomes one when the argument is first rounded up to the fixed dyadic grid $2^{-s}\mathbb{Z}$, i.e. replaced by $\lceil y\, 2^s \rceil / 2^s$ before the root is taken; the evaluation takes u from (13.4) applied to that grid point (with $d = 2^s$ fixed) at every step of Lemma 13.1's iteration, rounds the square root in (14.4) the same way, and takes $L^{\mp}$ and $t^{\mp}$ themselves as the least multiples of 2^{-s} at or above the values so obtained. Every such step is a composition of non-decreasing maps, and every rounding is upward, so (i) is unaffected;

(iii) *(half-width)* for $0 < p < 1$ and $\varepsilon^{\mp} > 0$, the guarded endpoints of R_{B} satisfy

$$0 \;\leq\; q^+(\hat{c}) - p \;\leq\; \frac{t^+}{n} + \frac{1}{n}, \qquad 0 \;\leq\; p - q^-(\hat{c}) \;\leq\; \frac{t^-}{n} + \frac{1}{n}, \tag{14.6}$$

where, for the unrounded value of (14.4),

$$\frac{t^{\mp}}{n} \;=\; \frac{L^{\mp}}{3n} + \sqrt{\Big(\frac{L^{\mp}}{3n}\Big)^{2} + \frac{2p(1-p)L^{\mp}}{n}} \tag{14.7}$$

is strictly decreasing in n and is $O(n^{-1/2})$ at fixed p and $\varepsilon^{\mp}$. The rule never emits an interval narrower than the raw quantile interval of the exact rule of Definition 8.1 (Proposition 8.2 (iii)).

15 The exact rule's local evaluation

Proposition 15.3 (The enclosure evaluation of the exact rule)**.** Assume a conservative evaluation (Definition 4.3). Let $n \geq 1$, let $p \in (0,1)$ and $\varepsilon^{+} \in (0,1)$ be rational, let N be binomial with parameters (n,p), and let κ^{+} be the exact upper quantile of Definition 8.1 at (n,p,ε^{+}). Let

$$\tau^{-}(k) \;\leq\; \Pr[\mathsf{N} > k] \;\leq\; \tau^{+}(k), \qquad k = 0,\dots,n, \tag{15.9}$$

be an enclosure with $\tau^{-}(n) = \tau^{+}(n) = 0$, each bound either a rational or given through a rational bound on its logarithm, such as the one Lemma 15.2 (ii)–(iii) yields with the mass point enclosed by Lemma 15.1 and the logarithms by Lemma 13.1. Call k *certified admissible*, $\mathsf{C}^{+}(k)$, when the available upper bound proves $\Pr[\mathsf{N} > k] \leq \varepsilon^{+}$: $\tau^{+}(k) \leq \varepsilon^{+}$, or in log space $U(k) \leq \lambda^{-}$ for rationals $U(k) \geq \ln \Pr[\mathsf{N} > k]$ and $\lambda^{-} \leq \ln \varepsilon^{+}$; and *certified inadmissible*, $\mathsf{D}^{+}(k)$, when the lower bound proves $\Pr[\mathsf{N} > k] > \varepsilon^{+}$: $\tau^{-}(k) > \varepsilon^{+}$, or in log space $\Lambda(k) > \lambda^{+}$ for rationals $\Lambda(k) \leq \ln \Pr[\mathsf{N} > k]$ and $\lambda^{+} \geq \ln \varepsilon^{+}$. An index that is neither is *undecided*. The predicate $\mathsf{C}^{+}(n)$ always holds.

A *bracketed upper endpoint* is an integer $\tilde{k}^{+} \in \{0,\dots,n\}$ with $\mathsf{C}^{+}(\tilde{k}^{+})$ and, if $\tilde{k}^{+} \geq 1$, $\neg\mathsf{C}^{+}(\tilde{k}^{+}-1)$. The *enclosure evaluation of the exact rule* is the endpoint rule R_{encl} whose R^{+}_{encl} is the bracketed upper endpoint returned by a fixed deterministic search procedure that returns such an endpoint for every input, whose R^{-}_{encl} is obtained in the same way for the lower tail through the mirror $\mathsf{N}' = n - \mathsf{N}$ of Lemma 15.2 (v), and which returns the values of Definition 7.1 (b) and (c) at a zero budget and at $p \in \{0,1\}$. Then:

(i) *(admissible relaxation)* Every bracketed upper endpoint satisfies $\Pr\big[\mathsf{N} > \tilde{k}^{+}\big] \leq \varepsilon^{+}$ and $\kappa^{+} \leq \tilde{k}^{+}$; mirror-wise every bracketed lower endpoint $\tilde{k}^{-}$ satisfies $\Pr\big[\mathsf{N} < \tilde{k}^{-}\big] \leq \varepsilon^{-}$ and $\tilde{k}^{-} \leq \kappa^{-}$. Hence R_{encl} is an admissible endpoint rule, Lemma 7.2 applies to it, and the box $\mathsf{Acc}_{R_{\mathrm{encl}}}$ emits (Definition 9.1) contains the box the exact rule emits: every q^{-} is at most and every q^{+} at least the exact rule's.

(ii) *(the bracket decided)* If $\tilde{k}^{+} \geq 1$ and $\mathsf{D}^{+}(\tilde{k}^{+}-1)$ holds, then $\tilde{k}^{+} = \kappa^{+}$; if $\tilde{k}^{+} = 0$ then $\tilde{k}^{+} = \kappa^{+}$. The mirror statement holds for the lower side.

(iii) *(conservatism in counts)* Suppose $\tilde{k}^+ > \kappa^+$, $\kappa^+ \le n-2$ and $r_{\kappa^+ +1} < 1$, with r_j the ratios (15.4). Let $w \ge 0$ be the slack of the failed certificate at $\tilde{k}^+ - 1$: $w := \ln \tau^+(\tilde{k}^+ - 1) - \ln \Pr\left[\mathsf{N} > \tilde{k}^+ - 1\right]$, or in log space $w := U(\tilde{k}^+ - 1) - \ln \Pr\left[\mathsf{N} > \tilde{k}^+ - 1\right] + (\ln \varepsilon^+ - \lambda^-)$. Then

$$\tilde{k}^+ - \kappa^+ \;<\; 1 + \frac{w}{-\ln r_{\kappa^+ +1}}. \tag{15.10}$$

The emitted endpoint is therefore the exact quantile up to a number of counts bounded by the enclosure's slack divided by the per-count decay of the tail at the quantile. The mirror statement holds for the lower side, through N' (Lemma 15.2 (v)): with $\kappa^- = n - \kappa'^+$ and $\tilde{k}^- = n - \tilde{k}'^+$, where the primed quantities are those of N' at level ε^-, $\kappa^- - \tilde{k}^- < 1 + w'/(-\ln r'_{\kappa'^+ +1})$ under the mirrored hypotheses. Consequently, wherever the bracket is decided the emitted endpoint is the exact rule's, and so are its guarded endpoint and its box bound; wherever it is undecided the guarded endpoint exceeds the exact rule's by at most $\tilde{k}^+ - \kappa^+$ counts and the box bound q^+ the exact rule's by at most $(\tilde{k}^+ - \kappa^+)/n$, which (15.10) bounds under the hypotheses above; mirror-wise for k^- and q^-. Where the hypothesis $r_{\kappa^+ +1} < 1$ fails (the exact quantile at or below the mean, reachable at a near-unit tail budget, Proposition 8.2 (vi)) the endpoint is still admissible by (i) and only this bound is unavailable; at $\kappa^+ = n - 1$ the excess $\tilde{k}^+ - \kappa^+$ is at most one count.

16 Requirements: the request, its refusal and the reports

Requirement 16.1 (The rule the request selects). The engine builds the box of Definition 9.1 with the rule that the request's `endpoint_rule` and `tail_evaluation` select, and with no other: `endpoint_rule:exact` with `tail_evaluation:sweep`, the exact binomial quantile rule of Definition 8.1 with its tail comparisons decided on the closed forms, exactly or against upper bounds on the tails as Proposition 8.2 (vii) admits, which is the default; `endpoint_rule:exact` with `tail_evaluation:enclosure`, the enclosure evaluation R_{encl} of the exact rule (Proposition 15.3), an admissible relaxation of it; and `endpoint_rule:bernstein`, the Bernstein rule of Proposition 14.2, on which `tail_evaluation` selects only how the completeness figure is computed (Requirement 16.4).

Requirement 16.2 (Defaults for the constrained outcomes and the allocation). Where a request omits the constrained outcomes, the engine takes $\widehat{\mathcal{C}}_{\mathrm{con}} = \widehat{\mathcal{C}}$ (`constrained_outcomes`); where it omits the allocation, the

engine takes the uniform rule of Definition 12.1 at the request's budget, $\varepsilon^{\pm}(\hat{c}) = \varepsilon_{\mathrm{comp}}/(2|\widehat{\mathcal{C}}_{\mathrm{con}}|)$ for every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{con}}$ (`budget_allocation`).

Requirement 16.3 (Refusal of an input that violates a decidable clause). Before it constructs anything, the engine decides exactly every condition of an admissible input (Definition 7.1) that its own inputs determine on data of the form that definition fixes. On a violation it emits `infeasibility_report` naming the violated condition by its code, the first violation in the order the interface fixes, and it emits no `acceptance_set`; it repairs no input and puts no default in place of a violating value. The codes of the violations (1)–(11) of Theorem 11.1 (iii) are, in that order, `alphabet_empty`, `alphabet_not_finite`, `rounds_out_of_range`, `honest_not_a_distribution`, `honest_envelope_not_admissible`, `budget_out_of_range`, `allocation_negative`, `allocation_exceeds_budget`, `selector_not_a_subset`, `inputs_not_rational` and `endpoint_rule_unknown`. Data that, with the defaults of Requirements 16.2 and 16.1 in place, do not have that form are refused as well: the engine emits no `acceptance_set` for them and substitutes nothing for them.

Requirement 16.4 (The reported completeness figure). The engine reports under `completeness_certified` the following figure. On `endpoint_rule:exact` with `tail_evaluation:sweep` it reports $\varepsilon^{\star}$ of Theorem 10.2 itself, and where it cannot evaluate a tail exactly it rounds the tail upward and reports $\min\{\varepsilon_{\mathrm{comp}}, \tilde{\varepsilon}\}$ for the resulting $\tilde{\varepsilon} \geq \varepsilon^{\star}$ (Remark 10.3). On every other route it reports the admissible figure of Remark 14.3 that the route's row of `completeness_report` names: the allocation sum, or $\min\{\varepsilon_{\mathrm{comp}}, \tilde{\varepsilon}\}$ for an upward-rounded evaluation $\tilde{\varepsilon} \geq \varepsilon^{\star}$ of the tails at the emitted endpoints. It never reports a downward-rounded figure.

Requirement 16.5 (The honest-model record beside every completeness figure). With every completeness figure it reports, the engine records, in the record `honest_model`, the input and the evaluation under which the figure was computed: the honest model (the law in the point form, or the honest envelope), the round count n, the constrained outcomes $\widehat{\mathcal{C}}_{\mathrm{con}}$ and the allocation; the endpoint rule and the tail evaluation; and the parameters in force, for the Bernstein rule its j and s (Proposition 14.2) and, wherever the tails are evaluated by enclosures, the parameters of those enclosures (Proposition 15.3). A completeness figure is meaningful only relative to that record.

17 Requirements: the enclosure evaluation

Requirement 17.1 (One fixed search for the enclosure evaluation). On the enclosure evaluation of the exact rule, the engine finds every bracketed

endpoint of Proposition 15.3, upper and lower, by one fixed search procedure, total and deterministic, with fixed parameters: the procedure `search_procedure` records.

REQUIREMENT 17.2 (Log-space arithmetic with directed rounding). On the enclosure evaluation the engine computes everything in log space with directed rounding: the mass point through (15.2), the window through the rationals A_T and B_T of (15.6), whose logarithms Lemma 13.1 encloses, and the budget through the same lemma. It holds the products $\prod r_l$ inside A_T and B_T as dyadic numbers with an explicit binary exponent, rounded in the direction each bound requires (down for A_T in the lower bound, up for $A_T + B_T$ in the upper), not as exact rationals, and the Stirling remainders and the root bounds as rationals of the size of a logarithm; the only numbers it ever forms have the magnitude of logarithms.

REQUIREMENT 17.3 (The bracket verdict per outcome and side). On the enclosure evaluation of the exact rule the engine records in the diagnostic `endpoint_exact`, per constrained outcome and per side, whether the bracket was decided, that is whether the emitted endpoint is proved equal to the exact quantile by Proposition 15.3 (ii). When it holds a certified-inadmissible index k_{D} below the bracket with $0 < r_{k_{\mathrm{D}}+2} < 1$, it may also report the bound (15.10) in counts with $-\ln r_{k_{\mathrm{D}}+2}$ in place of $-\ln r_{\kappa^+ +1}$ and the computable upper bound $U(\tilde{k}^+ - 1) - \Lambda(\tilde{k}^+ - 1) + (\lambda^+ - \lambda^-)$ on w in place of w (Remark 15.4).

Singleton acceptance set

18 Completeness of a singleton acceptance set

Theorem 18.1 (Completeness of a singleton acceptance set). Assume A-HON, let $\widehat{\mathcal{C}}$ and n be a matched alphabet and round count (Definition 1.3), and let $\mathcal{S}_{\mathrm{acc}} = \{\mathbf{q}^*\}$ be a singleton geometry with $\mathbf{q}^* \in \Delta(\widehat{\mathcal{C}})$. Call a completeness budget $\varepsilon_{\mathrm{comp}}$ *admissible for this geometry* when $\varepsilon_{\mathrm{comp}} \in [0, 1)$ and $\Pr\left[\mathrm{freq}_{\widehat{C}_1^n} \notin \mathcal{S}_{\mathrm{acc}}\right] \le \varepsilon_{\mathrm{comp}}$.

(i) *(exact accept probability)* With the convention $0^0 = 1$,

$$\Pr\left[\mathrm{freq}_{\widehat{C}_1^n} \in \mathcal{S}_{\mathrm{acc}}\right] = \begin{cases} \dfrac{n!}{\prod_{\hat{c}\in\widehat{\mathcal{C}}} (n\,q^*(\hat{c}))!} \displaystyle\prod_{\hat{c}\in\widehat{\mathcal{C}}} p^{\mathrm{hon}}(\hat{c})^{\,n\,q^*(\hat{c})}, & \text{if } n\,q^*(\hat{c}) \in \mathbb{Z}_{\ge 0}\ \forall \hat{c}, \\ 0, & \text{otherwise;} \end{cases} \tag{18.1}$$

(ii) *(the least admissible budget, and when there is none)* the accept prob-

ability (18.1) is 0 if and only if either $n\,q^*(\hat{c}) \notin \mathbb{Z}_{\geq 0}$ for some $\hat{c}$, or $\mathbf{q}^* \not\ll p^{\mathrm{hon}}$, that is, some $\hat{c}$ has $q^*(\hat{c}) > 0$ and $p^{\mathrm{hon}}(\hat{c}) = 0$. In that case *no* $\varepsilon_{\mathrm{comp}} \in [0, 1)$ is admissible for this geometry, and no completeness statement whatsoever can be made about it; a budget of 1 lies outside $[0, 1)$. Otherwise the accept probability is strictly positive and the least admissible completeness budget is

$$\varepsilon_{\mathrm{comp}} = 1 - \Pr\left[\,\mathrm{freq}_{\widehat{C}_1^n} \in \mathcal{S}_{\mathrm{acc}}\right] \;\in [0, 1), \tag{18.2}$$

which does lie in $[0, 1)$;

(iii) *(when the budget is zero)* the budget (18.2) equals 0 if and only if p^{hon} is a point mass $\delta_{\hat{c}_0}$ for some $\hat{c}_0 \in \widehat{\mathcal{C}}$ and $\mathbf{q}^* = \delta_{\hat{c}_0}$; in particular it equals 0 whenever $|\widehat{\mathcal{C}}| = 1$, and it is strictly positive whenever p^{hon} gives positive mass to two distinct outcomes.

19 Requirements

Requirement 19.1 (The completeness budget of a declared singleton). An instance that declares a singleton acceptance set $\{\mathbf{q}^*\}$ records the completeness budget $\varepsilon_{\mathrm{comp}}$ of (18.2), or, in the degenerate case of Theorem 18.1 (ii) where the accept probability is exactly 0, records that *no* admissible completeness budget exists rather than a budget of 1, which is the honest reading of a geometry the honest run can never satisfy. In either case it claims no completeness the set does not have.

Requirement 19.2 (Point-form honest input for the singleton). The singleton takes its honest input in the point form only: the engine reads a degenerate envelope as its law, and refuses a non-degenerate envelope with `honest_envelope_not_degenerate` rather than reducing it to one of its laws.

References